

 PharmaEZ — Pharmacy & Healthcare E-Commerce

Phase	Project Design Phase
Document	Proposed Solution
Project Name	PharmaEZ — Pharmacy & Healthcare E-Commerce
Team ID	[TEAM-ID]
Date	14 June 2026
Maximum Marks	2 Marks

Proposed Solution Template

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1	Problem Statement (Problem to be solved)	Primary Problem: There is no single, trustworthy, and user-friendly online pharmacy platform that seamlessly handles prescription-based medicine purchases (with proper pharmacist verification of Rx uploads), OTC product discovery with complete clinical metadata, and provides pharmacy owners with an integrated management and analytics dashboard — all within one MERN stack application. Key Pain Points Addressed: ● Patients cannot buy Rx medicines online without a clear, compliant prescription upload workflow ● Healthcare product pages on general marketplaces lack salt composition, dosage form, and side-effects information ● No affordable all-in-one solution for small pharmacy owners to manage inventory, orders, and prescriptions digitally ● Low trust in counterfeit products on general e-commerce platforms ● No loyalty or engagement mechanism to drive repeat healthcare purchases
2	Idea / Solution Description	PharmaEZ — Pharmacy & Healthcare E-Commerce Web Application A full-stack MERN (MongoDB, Express.js, React.js, Node.js) e-commerce web application purpose-built for the pharmacy and healthcare domain. PharmaEZ enables customers to browse, search, and purchase prescription medicines, OTC drugs,

		vitamins, medical devices, and wellness products through a responsive, mobile-first web interface. Core Modules Delivered: • JWT-secured user authentication with role-based access (user / admin) • Pharmacy product catalogue with 10 categories and full clinical metadata (salt composition, dosage form, strength, side effects, usage instructions) • Full-text search with advanced filters (category, price range, Rx required) and pagination • Prescription (Rx) upload workflow with pharmacist review, approve/reject, and status tracking • Persistent shopping cart with real-time stock validation and free-shipping threshold logic • 3-step checkout: Address → Payment Method → Order Review • Order management with 5-step tracking (placed → confirmed → processing → shipped → delivered) and full audit trail • Product reviews and star ratings with aggregate recalculation • Loyalty points system (1 point per ₹10 spent) • Admin dashboard with KPIs, 7-day revenue chart, order management, inventory alerts, user management, and prescription review
3	Novelty / Uniqueness	• Healthcare-Specific Product Metadata: Unlike general e-commerce platforms, PharmaEZ displays salt composition, generic name, dosage form, strength, side effects, and usage instructions on every product page — enabling informed, safe medicine decisions. • Integrated Rx Compliance Workflow: The prescription upload → pharmacist review → order link workflow is built into the core purchase flow — not bolted on as an afterthought. Rx items are clearly flagged and cannot be purchased without prescription verification. • Dual-Role Platform: A single application serves both end consumers (with a clean shopping experience) and pharmacy administrators (with a full management dashboard) — eliminating the need for separate systems. • Loyalty Points for Healthcare: A 1-point-per-₹10 loyalty programme drives repeat purchases, which is critical for the high-frequency, recurring nature of medicine purchases. • MERN Stack Architecture: Clean MVC separation, JWT authentication, Context API state management, and Mongoose schema validation make PharmaEZ production-ready and scalable without requiring proprietary cloud services.

4	Social Impact / Customer Satisfaction	- Improves healthcare access for patients in cities and semi-urban areas by enabling doorstep medicine delivery - Reduces risk of purchasing counterfeit or expired medicines through genuine-product verification and pharmacist oversight - Empowers elderly patients and caregivers who cannot easily visit physical pharmacies - Promotes prescription compliance — the Rx workflow encourages patients to obtain and share proper prescriptions - Supports small pharmacy owners with affordable digital tools to compete with large pharmacy chains - Reduces wasted trips to pharmacies for out-of-stock medicines through real-time inventory display - Customer satisfaction indicators: loyalty points reward repeat customers; reviews and ratings build informed product choices
5	Business Model (Revenue Model)	Multiple revenue streams are planned for PharmaEZ: Commission on Transactions: Platform commission (5–10%) on every order placed through PharmaEZ. Pharmacy Subscription Plans: Monthly/annual subscription for pharmacy admins to access the full management dashboard, inventory tools, and analytics. Premium Product Listings: Pharmaceutical manufacturers pay for featured placement on homepage banners and category pages. Delivery Charges: Flat ₹49 delivery fee on orders below ₹499 free-delivery threshold. Loyalty Programme Monetisation: Sponsored loyalty point top-ups and redemptions offered by partner brands. Data Insights (Future): Anonymised purchase trend reports offered to pharmaceutical distributors for inventory planning.
6	Scalability of the Solution	Horizontal Scalability: The Node.js backend is stateless (JWT-based authentication); additional instances can be deployed behind a load balancer to handle increased traffic. Database Scalability: MongoDB's document model supports flexible schema evolution. Sharding and replica sets can be configured for horizontal database scaling as the product catalogue and order volumes grow. Microservices Migration Path: The current MVC monolith is structured modularly (separate controllers for auth, products, cart, orders, admin) — enabling decomposition into independent microservices (Prescription Service, Order Service, Inventory Service) as traffic demands increase. Multi-Pharmacy / Multi-Vendor: The data model includes a seller/manufacture reference on the Product schema —

		enabling a multi-vendor marketplace expansion where multiple pharmacies list products on the platform. ● Multi-City / Pan-India: Adding city-based delivery zone configuration and pin-code serviceability checks requires only backend logic additions — the frontend filter system already supports it. ● Cloud-Ready: The application is designed for deployment on AWS (EC2 + MongoDB Atlas) or Heroku/Railway with environment-variable-driven configuration — zero hardcoded infrastructure dependencies.
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